Course Syllabus

ECON 6130: Macroeconomics I (2nd half)

Fall 2026

Course Homepage

Time and Place:
Lecture: Monday and Wednesday (first meeting October 19), 10:10am - 11:25am, Ives Hall 111
Discussion: Friday, 8:40am - 9:55am, Goldwin Smith Hall 142

Professor:
Ryan Chahrour
Email: ryan.chahrour@cornell.edu
Office hours: Mon @ 4:30pm-5:30pm and by online appointment (Uris 428)

Teaching Assistant:
Ayush Mathur
am3522@cornell.edu
Office hours: Mon @ 4:30pm-5:30pm and Tue @ 1:30pm-2:30pm (Uris 447)

Course Overview:

This is a course on basic macroeconomics, but with a strong focus on the computational issues involved in solving and evaluating macroeconomic models. The techniques taught in the course will be useful to any applied quantitative researcher, regardless of field.

The course first introduces a variant of the real business cycle model, with the addition of a search friction in the labor market. We use this model to teach the techniques and concepts that can be used to solve virtually any representative-agent model in macroeconomics. Along the way, we provide a general introduction to numerical computing, including root finding, numerical optimization, numerical differentiation, numerical integration, and functional approximation.

To solve our example economy, we introduce and implement four computational solution strategies that have wide application: linearization, the shooting method, value function iteration, and the projection technique. Numerical stability, quality of approximation, and speed are assessed for these methods in the context of our labor search model.

Time permitting, we will look at how these tools can be extended to a variety of other contexts, including models of forward economic news, models of the interaction between the macroeconomy and the climate, and macro models of heterogeneous agents.

Coursework will consist of four problem sets (30%) and an in-class final exam on December 7 (70%). *Problem sets should be turned in; however, problem set grades will be based primarily on brief quizzes in course discussion sections.* To complete the problem sets, you should have access to a computer and a recent version of MATLAB (available free under Cornell's site license).

Optional Text:

- Mario J Miranda and Paul L Fackler. *Applied Computational Economics and Finance*. MIT Press, 2004

Useful background material on computation:
https://www.chahrour.net/teaching/numerical-methods-for-economists

Statement regarding AI/LLMs:
AI tools are useful for economists, and you should build your comfort with them as a graduate student. I use AI tools to help organize the course materials, for feedback on course design choices, and to check my work. I will not share with you any AI-generated material that I have not personally verified.

As a student in this course, *you should use AI with extreme caution.* Using these tools to support your understanding of a concept or a paper is fine. You may also use AI as an aid with problem sets, but I encourage you to avoid this as much as possible. The more you rely on AI for this work, the more you risk missing key ideas and learning opportunities. If a quiz reflects incomplete understanding of homework material and your submitted work appears to be AI-generated, we may assign an additional penalty. If you wish to check algebra, I recommend using MATLAB's Symbolic Math Toolbox, which is available for free as part of Cornell's MATLAB site license.

Tentative Topic/Lecture Outline

Topic 1: A labor search model (with numerical introductions)

- Robert Shimer. The Cyclical Behavior of Equilibrium Unemployment and Vacancies. *The American Economic Review*, 95(1):25–49, 2005
- Robert E. Hall. Employment Fluctuations with Equilibrium Wage Stickiness. *The American Economic Review*, 95(1):50–65, 2005
- Marcus Hagedorn and Iourii Manovskii. The Cyclical Behavior of Equilibrium Unemployment and Vacancies Revisited. *American Economic Review*, 98(4):1692–1706, September 2008
- Ryan Chahrour, Sanjay K Chugh, and Tristan Potter. Anticipated Productivity and the Labor Market. *Quantitative Economics*, 14(3):897–934, 2023

Topic 2: Perturbation techniques

- Paul Klein. Using the Generalized Schur Form to Solve a Multivariate Linear Rational Expectations Model. *Journal of Economic Dynamics and Control*, 24:1405–1423, 2000
- Stephanie Schmitt-Grohé and Martín Uribe. Solving Dynamic General Equilibrium Models Using a Second-Order Approximation to the Policy Function. *Journal of Economic Dynamics and Control*, 28(4):755–775, 2004
- Dario Caldara, Jesus Fernandez-Villaverde, Juan F Rubio-Ramirez, and Wen Yao. Computing DSGE Models with Recursive Preferences and Stochastic Volatility. *Review of Economic Dynamics*, 15(2):188–206, 2012

Topic 3: The perfect-foresight shooting method

- Jeremy Greenwood and Ricardo Marto. Numerical Methods for Macroeconomists (with Julia and Matlab Codes). Manuscript, 2022

Topic 4: Interpolation and extrapolation

- Mario J Miranda and Paul L Fackler. *Applied Computational Economics and Finance*. MIT Press, 2004

Topic 5: Value function iteration

- S. Borağan Aruoba, Jesús Fernández-Villaverde, and Juan F. Rubio-Ramírez. Comparing Solution Methods for Dynamic Equilibrium Economies. *Journal of Economic Dynamics and Control*, 30(12):2477–2508, 2006

Topic 6: Projection methods

- Alexander W. Richter, Nathaniel A. Throckmorton, and Todd B. Walker. Accuracy, Speed and Robustness of Policy Function Iteration. *Computational Economics*, pages 1–32, 2013

Topic 7: Empirical methods in macro

- Christopher A. Sims. Macroeconomics and Reality. *Econometrica*, 48(1):1–48, 1980

- Lawrence J Christiano. Christopher A. Sims and Vector Autoregressions. *The Scandinavian Journal of Economics*, 114(4):1082–1104, 2012

Topic 8: Business cycles and macroeconomic news

- Robert J. Barro and Robert G. King. Time-Separable Preferences and Intertemporal-Substitution Models of Business Cycles. *Quarterly Journal of Economics*, 99(4):817–839, 1984

- John H Cochrane. Shocks. In *Carnegie-Rochester Conference Series on Public Policy*, volume 41, pages 295–364. Elsevier, 1994

- Paul Beaudry and Franck Portier. Stock Prices, News, and Economic Fluctuations. *The American Economic Review*, 96(4):1293–1307, 2006

- Nir Jaimovich and Sergio Rebelo. Can News about the Future Drive the Business Cycle? *American Economic Review*, 99(4):1097–1118, 2009

- Ryan Chahrour, Sanjay K Chugh, and Tristan Potter. Anticipated Productivity and the Labor Market. *Quantitative Economics*, 14(3):897–934, 2023

Topic 9: Climate change in macroeconomics

- Mikhail Golosov, John Hassler, Per Krusell, and Aleh Tsyvinski. Optimal Taxes on Fossil Fuel in General Equilibrium. *Econometrica*, 82(1):41–88, 2014

Topic 10: Neural networks and machine learning in macro

- Lilia Maliar, Serguei Maliar, and Pablo Winant. Deep Learning for Solving Dynamic Economic Models. *Journal of Monetary Economics*, 122:76–101, 2021

- Mahdi Ebrahimi Kahou, Jesús Fernández-Villaverde, Sebastian Gomez Cardona, Jesse Perla, and Jan Rosa. Spooky Boundaries at a Distance: Inductive Bias, Dynamic Models, and Behavioral Macro. Technical report, CEPR Discussion Papers, 2024

- Jesús Fernández-Villaverde, Galo Nuño, and Jesse Perla. Taming the Curse of Dimensionality: Quantitative Economics with Deep Learning. Technical report, National Bureau of Economic Research, 2024

Topic 11: Heterogeneous agent macroeconomics

- S. Rao Aiyagari. Uninsured Idiosyncratic Risk and Aggregate Saving. *The Quarterly Journal of Economics*, 109(3):659–684, August 1994

- P. Krusell and A.A. Smith, Jr. Income and Wealth Heterogeneity in the Macroeconomy. *Journal of Political Economy*, 106(5):867–896, 1998

- Sushant Acharya and Keshav Dogra. Understanding HANK: Insights from a PRANK. *Econometrica*, 88(3):1113–1158, 2020